

Student Feedback Sentiment Classification Using Sentence Embeddings

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1 Introduction

This project addresses a text classification problem in the educational domain: classifying student feedback as *negative*, *neutral*, or *positive*. Universities collect feedback about teaching, course content, examinations, lab work, library facilities, and extracurricular activities. Manually reviewing this feedback is time-consuming, especially when negative or neutral comments need to be identified. The project uses sentence embeddings to represent feedback comments as numerical vectors and trains supervised classifiers on top of these embeddings.

2 Dataset

The dataset was created from a Kaggle student feedback spreadsheet. The raw file contained paired columns for several feedback areas: one numeric sentiment label and one free-text student comment. I processed this into a long-format text classification dataset where each row represents one feedback comment with the columns `text`, `category`, `sentiment_score`, and `sentiment`. The final dataset contains 724 rows. A manual sanity check was performed to verify that the sentiment labels generally matched the feedback text.

Class	Examples	Share
Positive	451	62.3%
Negative	139	19.2%
Neutral	134	18.5%

Table 1: Class distribution in the cleaned dataset.

3 Method

The project uses `sentence-transformers/all-MiniLM-L6-v2` to convert each feedback comment into a dense sentence embedding. Embeddings are useful because student feedback is often short, informal, and semantically varied. A keyword-based method may miss comments such as “needs improvement” or “good but limited”, while sentence embeddings can capture semantic similarity beyond exact word matching.

The dataset was split into 80% training data and 20% test data, stratified by sentiment label. Three classifiers were trained on the same embeddings and evaluated on the same held-out test set: Logistic Regression, Linear SVM, and Random Forest. Class weighting was used where applicable to reduce the effect of class imbalance.

4 Results

The models were evaluated on a held-out test set of 145 examples. The results are shown in Table 2.

Classifier	Accuracy	Macro F1	Weighted F1
Logistic Regression	0.690	0.612	0.712
Linear SVM	0.752	0.609	0.729
Random Forest	0.676	0.447	0.597

Table 2: Classifier performance on the test set.

Linear SVM achieved the highest accuracy and weighted F1-score, but Logistic Regression achieved the highest macro F1-score. Macro F1-score was used as the main selection metric because the dataset is imbalanced, with more positive examples than neutral or negative examples. Logistic Regression was therefore selected for the Hugging Face demo, as it gave the best balance across the three sentiment classes. Random Forest performed worse, especially on macro F1, suggesting that it struggled more with the minority classes.

5 Web Demo and Repository

The solution is implemented as a Gradio demo for Hugging Face Spaces. The user enters a student feedback comment and the demo returns predicted sentiment probabilities.

- GitHub repository: <https://github.com/hanneslindqvist7265/Information-retrieval-assignment>
- Hugging Face dataset: https://huggingface.co/datasets/Hannes111/student_feedback_sentiment
- Hugging Face demo: <https://huggingface.co/spaces/Hannes111/student-feedback-sentiment>

6 Reflection on Using AI Tools

AI tools were used for coding support, dataset processing, project structure, and documentation. This made the development process faster, especially when creating the training pipeline and Gradio demo. However, the outputs still needed to be checked carefully. The dataset labels were manually inspected, and the code and evaluation results were verified before being used. This assignment showed that AI tools can be useful programming assistants, but the responsibility for understanding and validating the final work remains with the user.

7 Conclusion

This project shows how sentence embeddings can be used for student feedback sentiment classification. A custom dataset was created by processing raw spreadsheet data into a text classification format. Several classifiers were trained and evaluated, and Logistic Regression was selected for the final Hugging Face demo based on macro F1-score. Future work could include expanding the dataset, improving label quality, and analyzing sentiment separately for each feedback category.

References

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- [2] Hugging Face. Models, Datasets, and Spaces. <https://huggingface.co/>
- [3] Kaggle. Student Feedback Dataset. <https://www.kaggle.com/datasets/brarajit18/student-feedback-dataset>